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Jacques Brisson, Yves Bergeron, André Bouchard & Alain Leduc

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Beech-maple dynamics in an old-growth forest in southern Québec, Canada¹

Jacques BRISSON, Institut de recherche en biologie végétale, 4101 est, rue Sherbrooke, Montréal, Québec H1X 2B2, Canada., Fax: (514) 987-4872

Yves BERGERON, GREF, Université du Québec à Montréal, C.P. 8888, Succ. A, Montréal, Québec H3C 3P8, Canada.

André BOUCHARD, Institut de recherche en biologie végétale, 4101 est, rue Sherbrooke, Montréal, Québec H1X 2B2, Canada.

Alain LEDUC, GREF, Université du Québec à Montréal, C.P. 8888, Succ. A, Montréal, Québec H3C 3P8, Canada..

Abstract: A study of forest succession was conducted in an 11 ha old-growth beech-maple stand located at the northern limit of the temperate deciduous forest of North America. Eventual changes in dominance were simulated from three transition matrix models based on understory composition or observed replacement in forest gaps. All the models suggested that the forest composition is not currently at equilibrium. If the transition tendency persists, American beech (*Fagus grandifolia* Ehrh.) will sharply increase in abundance at the expense of sugar maple (*Acer saccharum* Marsh.). Possible mechanisms responsible for this replacement are discussed.

Keywords: beech, maple, old-growth forest, succession, canopy gaps.

Résumé: Une étude de succession forestière a été entreprise dans une vieille érablière à hêtre de 11 ha, située à la limite nord de la forêt tempérée décidue d'Amérique du Nord. Les changements éventuels dans la dominance ont été simulés à l'aide de trois modèles de matrice de transition basés sur la composition botanique du sous-bois ou sur le remplacement observé dans les ouvertures du couvert forestier. Tous les modèles suggèrent que la composition forestière actuelle n'est pas à l'équilibre. Si les tendances au changement persistent, le hêtre d'Amérique (*Fagus grandifolia* Ehrh.) va fortement accroître son abondance aux dépens de l'érable à sucre (*Acer saccharum* Marsh.). Les mécanismes responsables de ce remplacement sont discutés.

Mots-clés: hêtre, érable, vieille forêt, succession, trouées.

Introduction

Most studies comparing the current tree composition of eastern North-American forests to presettlement composition have shown a strong decrease in the proportion of American beech (*Fagus grandifolia* Ehrh.) since European colonization, often coinciding with an increase in sugar maple (*Acer saccharum* Marsh.). For example, relative dominance of beech went from 50% to 13% in the Catskill Mountains, New York (McIntosh, 1972) and from 40% to 5% in northern Vermont (Siccama, 1971) with an increase in maple from 13% to 23% and 16% to 24%, respectively. In Ohio, beech dropped from 1st to 8th place in terms of relative abundance, while maple stayed in second place (Shanks, 1953). Similar tendencies have been observed in Wisconsin (Ward, 1956), Michigan (Wollet & Sigler, 1928), south-western New York (Gordon, 1940) and Pennsylvania (Lutz, 1930). Since the overall composition of the presettlement landscape is more likely to reflect a near equilibrium compared to present younger and more disturbed forests, this change could be interpreted as beech being more late-successional than maple. This hypothesis is supported by studies by Forcier (1975), Nicholson *et al.* (1979) and Leak (1987), who described maple as having a mid-successional status when compared with beech. However, present beech-maple forests recovering from disturbance do not exhibit a tendency to return to the presettlement dominance of beech over maple (McIntosh, 1972).

Recent interest in the dynamic relationship between beech and maple has focused on their response to gaps of different sizes (Runkle, 1990), the frequency of gap formation (Canham, 1988) or replacement patterns (Woods, 1979, 1984; Runkle, 1985). For example, it was found that maple has a lower capacity to withstand suppression (Canham, 1989; Poulson & Platt, 1989), but in general has a stronger growth response in the weak light of small gaps (Runkle, 1990; Canham, 1985, 1988). Also, each species does relatively better in gaps created by the other species (Woods, 1979, 1984). These functional differences have significant implications for the understanding of the species' dynamics. However, they do not fully account for the changes in relative abundance of maple and beech over time (Runkle, 1990) and the long-term successional relationship between these two species remains an open question.

An old growth beech-maple forest of presettlement origin was recently discovered in southern Québec (Brisson, Bergeron & Bouchard, 1988, 1992). A size-class ordination (Enright, 1982), using data collected in a 40 m × 40 m plot, suggested that beech dominance was increasing at the expense of maple (Brisson, Bergeron & Bouchard, 1988). In this paper, we have examined the dynamics of the dominant tree species using an approach based on a tree-by-tree replacement process. Three different methods were used to identify the species filling a gap. These analyses allowed us to determine if the current pattern of replacement will maintain the present forest composition.

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Study area

The study site was an 11 ha, old-growth forest located 70 km south-west of Montréal, in the Haut-Saint-Laurent region of Québec. This region is characterized by extensive lowlands covered with Champlain Sea clays. These lowlands are interrupted by morainic ridges lying parallel to the St. Lawrence River. Grandtner (1966) described the climax vegetation on mesic upland sites as a sugar maple and bitternut hickory (*Carya cordiformis* [Wang.] K.Koch.) association. Sugar maple dominates these forests, accompanied by bitternut hickory, basswood (*Tilia americana* L.), beech, ironwood (*Ostrya virginiana* [Mill.] K.Koch.) and white ash (*Fraxinus americana* L.). Disturbed forests on the uplands are dominated by gray birch (*Betula populifolia* Marsh), trembling aspen (*Populus tremuloides* Michx.) and large-toothed aspen (*Populus grandidentata* Michx.). The clay plain has been almost entirely deforested for agricultural use, while morainic deposits still support forest communities. These forests, generally covering small areas, have suffered anthropogenic disturbances of varying intensity and nature since European settlement some 200 years ago (Bergeron, Bouchard & Leduc, 1988).

The old-growth study site is located on a morainic ridge. Two vegetation types, related to hydric regime, were previously recognized (Brisson, Bergeron & Bouchard, 1992). The present study was conducted in the mesic portion, which is occupied by a beech-maple forest. In this portion, drainage varies from moderate to good. Maximum slope reaches 10%, although slopes are generally less than 5%. The soil is a brown stony loam with a pH ranging from 6.1 to 7.0 in the B horizon. Accompanying the maple and beech are ironwood, basswood and hemlock (*Tsuga canadensis* [L.] Carr.). The larger canopy trees range in age from 150 to 300 years. The understory is typical of a beech-maple forest, with a low percentage of ground cover, and the presence of acidophilic species such as *Polystichum acrostichoides* (Mx.) Schott and *Maianthemum canadense* Desf.

The lots on which the forest is located have belonged to the Muir family since its settlement in the 1820's. Because the first Muir had an unusual lack of interest in exploiting the wood (Sellar, 1888) and the subsequent family members have had a very conservative attitude, the forest has been exceptionally well protected from major human disturbances; only restricted partial cuts have occurred. Furthermore, there were no signs of maple dieback or beech bark disease at the time of sampling. At present, large dead trees are scattered throughout the forest, as snags or fallen logs with intact or highly decomposed wood. Since at least the seventeenth century, establishment and recruitment have depended on a gap-phase regeneration regime. A more complete description of the vegetation and history of the forest can be found in Brisson, Bergeron & Bouchard (1992).

Methods

The forest tree stratum was investigated using a grid of 20 × 20 m contiguous plots covering the entire mesic part of the forest. Every tree (diameter at breast height [d.b.h.] > 15 cm) was identified, mapped, and had its d.b.h. measured. Tree species that rarely reach the canopy, like ironwood,

were not considered as trees, even when they reached 15 cm in d.b.h.. A set of 112 beech saplings was sampled and excavated to determine if their origin was from seedling or root sprout.

Only four species, representing 93.7% of the total tree composition (relative density), were sufficiently abundant to be considered in the model: sugar maple (51.7%), American beech (23.4%), hemlock (9.3%) and basswood (9.3%). Rejected species included bitternut hickory (2.8%), white elm (*Ulmus americana* L.) (1.3%), yellow birch (*Betula alleghaniensis* Britton) (1.0%) and butternut (*Juglans cinerea* L.) (0.7%). Since these species were not abundant in the canopy nor in the understory, their inclusion in the model would often have resulted in small and inaccurate transition probabilities. However, the overall results for the four major species would have remained unaffected by such an inclusion. Therefore, the abundance of the four principal species was weighted to equal total forest composition in the model.

The model used here is based on a tree-by-tree replacement process. This approach is reasonably applicable to the mesic portion of the study site since openings in the canopy of the Muirs' forest appear to be the results of well interspersed dead trees rather than larger-scale gaps (Brisson, Bergeron & Bouchard, 1992). To estimate changes in tree composition, transition matrices P were used, entries p_{ij} being the probability that a tree of species j will replace one of species i . From the present forest composition, the composition of the next generation of trees can be calculated by:

$$n(t+1) = n(t)P$$

where $n(t)$ is a row vector ($n_1, n_2, \dots, n_k$) of the number of trees for each of the k species at generation t .

Whatever the composition of the starting vector, this type of model leads to the same stable composition after repeated iterations (regular Markov chain). Several authors have compared this stable equilibrium with the hypothesized climax for their study area, and have generally found close correspondence (Horn, 1975; Waggoner & Stephens, 1970). However, simple Markovian models assume that transition probabilities remain the same throughout succession, an assumption that is highly improbable (Binkley, 1980; Bellefleur, 1981; Facelli & Pickett, 1990). Models with time variations in the replacement probabilities have been developed (Acevedo, 1981), but the specific form of the function must be assumed. The application of simple stationary Markovian models is thus only meaningful for short term successional situations (Peet & Christensen, 1980), or for cases in which the model is based on vegetation already near a stationary state (Horn, 1975).

For the purpose of our study, only one iteration for each model was performed on present-day composition. The simulated composition represents the next generation of trees in the forest (*i.e.* when all trees presently in the canopy will be replaced), given the probabilities of replacement. This analysis gave an indication of the present direction of composition changes rather than the composition of a stable state. It also made it unnecessary to consider species' longevity when they were fairly comparable, as in this case.

Although this conservative use of transition matrices reduces the problems associated with the previously mentioned assumption, it does not circumvent those associated with the determination of the probabilities themselves. The building of transition probability matrices is a critical point of the approach, and the way they are calculated could lead to different predictions (White, MacKenzie & Busing, 1985). For the sake of comparison, we used three methods to calculate the matrices, based on different assumptions.

Model 1. This model is similar to that of Horn (1975). It assumes that: 1) any tree of the canopy will eventually be replaced by a suppressed sapling in its understory; and 2) the probability P_{ij} that an understory species j will replace an overstory i is proportional to the relative density of j in the understory of i . A sapling (d.b.h. < 15 cm; height > 1 m) was considered only when it was of the four species mentioned above, and when its crown was below the average canopy height. Every sapling located within 3 m of a tree was counted. A sapling could only be counted once, *i.e.* for the nearest tree. Sampling for this model was performed in 31 divisions of the forest grid (20% of total sampled area). A total of 1039 saplings under 281 trees were counted.

Model 2. With the same first assumption as in model 1, this model further assumes that any tree of the canopy will eventually be replaced by the largest sapling present in its understory. The probability P_{ij} becomes the relative number of times a sapling of species j is the largest one under a tree of species i . This model is based on which stem will capture the gap given the current size, with presumed greater probability given to the larger. This model further adds a spatial aspect compared with model 1. In the latter, computation could not differentiate between regularly distributed saplings and highly clumped saplings under a few trees. This situation would lead to unrealistic predictions. In model 2, only one sapling per tree is considered,

reducing the score for clumped saplings. Stem d.b.h. was used as a measure of size, since it is correlated with height, and much easier to measure. Other aspects of the sampling procedure for model 2 are the same as for model 1.

Model 3. Transition probabilities for model 3 were calculated from current replacement in the forest, by identifying current gap successors (Barden, 1979). We identified 114 dead or dying trees over the total area, and noted their apparent replacements, *i.e.* the individual stem that was occupying the largest area in the created gap, regardless of gap age or size (in every case but one, the stem occupying the largest area was also the tallest one in the gap). The probability of replacement became the relative number of times species j replaced species i in that manner.

Results

Based on the transition matrices (Table I), the three models predicted an important short term change in forest composition (Table II), implying that a disturbance regime exclusively regulated by a tree-by-tree replacement process, as simulated by the models, was insufficient to account for the present canopy composition. Moreover, predictions agreed strongly: all models predicted a sharp increase in beech over sugar maple.

Model 1 should have favored species that produce a large number of suppressed saplings waiting for an opening in the canopy, a strategy usually attributed to sugar maple. However, the model predicted a strong decrease in its relative abundance. In fact, maple regeneration is deficient or localized in the forest, despite an abundance of large seeding trees. It is particularly low under beech canopy, as shown by the transition probability: only 6.7% of saplings below beech are maple (Table I). The situation is better under maple trees, but even there, beech is abundant (42.7% for beech compared with 41.1% for maple). Overall, these two

TABLE I. Matrices of transition probabilities for the four major tree species in the Muirs' forest

Replaced tree (<i>i</i>)	Replacing stem (<i>j</i>)			
	<i>Acer saccharum</i>	<i>Fagus grandifolia</i>	<i>Tsuga canadensis</i>	<i>Tilia americana</i>
Model 1: From the relative density of saplings				
<i>Acer saccharum</i>	0.411 (+)*	0.427 (-)	0.142	0.019
<i>Fagus grandifolia</i>	0.067 (-)	0.873 (+)	0.060 (-)	0.000
<i>Tsuga canadensis</i>	0.351	0.324	0.324	0.000
<i>Tilia americana</i>	0.354	0.438	0.042	0.167 (+)
Model 2: From the largest saplings				
<i>Acer saccharum</i>	0.327	0.452	0.192	0.029
<i>Fagus grandifolia</i>	0.111 (-)	0.844	0.045 (-)	0.000
<i>Tsuga canadensis</i>	0.333	0.250	0.417	0.000
<i>Tilia americana</i>	0.450	0.200	0.050	0.300
Model 3: From replacement in gaps				
<i>Acer saccharum</i>	0.356	0.356	0.237	0.051
<i>Fagus grandifolia</i>	0.029	0.912	0.059	0.000
<i>Tsuga canadensis</i>	0.111	0.444	0.444	0.000
<i>Tilia americana</i>	0.000	0.182	0.091	0.727

* Conditional probabilities of replacement values (lines) significantly different from their unconditional probability (sum of table) were determined using Neu, Byers & Peek (1974) statistical deviations, calculated from the contingency tables (see Legendre & Legendre, 1983). Deviations were not calculated for cells of the contingency tables with values smaller than 5 (Hayes & Winkler, 1970). Significant differences ($p < 0.05$) are indicated by a symbol in parenthesis: (+), greater than expected; (-), smaller than expected.

species had significantly more saplings of their own species under their canopy than what was expected based on their respective number in the forest (Table I). The model further predicted an increase in hemlock, and a sharp decrease in the basswood population.

Results for model 2 were very similar. As in model 1, basswood was not the likely successor to beech, and maple and hemlock were significantly less likely to replace beech than was expected based on their number in the overall sample. The increase in hemlock was slightly stronger than predicted by model 1. This species produces fewer saplings but they usually have a better survival rate. These saplings consequently reach a larger size, accounting for hemlock's better score. Similar to model 1, the second model predicted a decrease in basswood's relative abundance. This situation could be the result of the violation of their first assumption of replacement by suppressed saplings. Basswood has a pronounced ability to produce vegetative sprouts that take advantage of the root system after the death of the main stem. The abundance (model 1) or size (model 2) of saplings under living trees became poor predictors of the future contribution of basswood to the forest canopy.

Model 3 did not depend on assumptions concerning probable replacement since it was directly developed from actual tree-by-tree replacement in the forest. However, as the number of dead mature trees was limited, we must note that the transition probabilities relied on fewer data. General trends were the same as in the other models, except that the basswood population was stabilized. The transition probabilities for this species show that dead basswood were mostly replaced by basswood (Table I), confirming the expectations mentioned in the first two models.

Discussion

Assuming a tree-by-tree replacement process, the models predicted a significant increase in dominance by beech, largely at the expense of maple. All our models used the current parameter structure and composition to estimate the future of the forest. White, MacKenzie & Busing (1985) discussed the hazards of using understory composition (models 1 and 2) to make predictions. Model 1 had several underlying assumptions that were questionable (Horn, 1975). Neither model 1, nor model 2 considered the species specific growth response to gaps. In model 3, gap size influence was not considered, a factor often mentioned as important in

subsequent composition (Runkle, 1985). It may thus be surprising that, despite their limitations, all these models based on different assumptions gave similar results regarding the dynamics of beech and maple. The prediction of maple replacement by beech was also supported by a previous study that used an approach related to demography in a small part of the forest (Brisson, Bergeron & Bouchard, 1988). We believe that this agreement between the analyses is less a result of the accuracy of the models than of the strength of this maple replacement tendency in the study site.

Although determination of the cause primarily responsible for the replacement must await further experimentation, several observations allow us to estimate the contributions of possibly important mechanisms. The expected shift in dominance can be examined as a result of two simultaneous mechanisms: the failure of maple to maintain its dominance and the success of beech in regeneration. The successional status of hemlock and basswood will be discussed later.

The most significant element that contributes to the change in composition is the failure of maple to regenerate abundantly under the present canopy. This is an uncommon situation for maple since it is usually a prolific shade-tolerant and self-perpetuating species, especially at this latitude (Grandtner, 1966). Maple maintains its dominance in mature forests by producing a large number of suppressed saplings that require an opening to be released (Canham, 1985). The mortality rate is usually very high during the seedling and sapling stages but progressively decreases with size, producing a diameter distribution as a steeply descending curve at equilibrium (Lorimer & Frelich, 1984). The deficiency in regeneration of maple in the study site is thus surprising given the omnipresence of seed-producing trees in the overstory. First year seedlings are observed to be abundant, but they rarely survive a full season (Brisson, Bergeron & Bouchard, 1988). In contrast, an adjoining forest stand on the same morainic ridge shows abundant maple regeneration (stand 1, Brisson, Bergeron & Bouchard, 1988). This stand is abiotically similar to the Muirs' forest but differs in its perturbation history: it has suffered repeated partial cuts. Thus, it is unlikely that the site's abiotic properties will satisfactorily explain the lack of maple regeneration. Maple grows well on mesic sites, and the present dominance of maple in the canopy of the Muirs' forest suggests that maple regeneration was very high at one time. Deer

TABLE II. Present and simulated future forest composition (relative density)* of the Muirs' forest calculated using transition matrices. Model 1, identity of gap-filling species calculated from the present relative density of saplings; model 2, from the larger saplings under canopy trees; model 3, from replacement in present gaps

	Present composition	Simulation model 1	Simulation model 2	Simulation model 3
<i>Acer saccharum</i>	55.2%	31.3%	28.6%	21.5%
<i>Fagus grandifolia</i>	25.0%	52.9%	50.5%	48.7%
<i>Tsuga canadensis</i>	10.0%	13.0%	16.4%	19.9%
<i>Tilia americana</i>	9.9%	2.7%	4.6%	10.0%

* Sum of the relative densities might not add up to 100% due to round off.

damage could favor beech over maple (Curtis & Rushmore, 1958). However, if grazing was significant enough to decrease maple regeneration, it would have damaged the adjacent forests as well.

The negative effect of some properties of the soil induced by the growing population of beech may cause the deficiency in maple's regeneration. In the Muirs' forest, like in most other old forests where beech is abundant, the understory vegetation is scanty and dispersed, especially under beech canopy (Rogers, 1982). It has long been shown that large differences exist between the nutrient content of beech litter compared with maple (Colkwell & DeLong, 1950; Scott, 1955). For example, beech litter has a high C:N content, a low pH (Bouliane, 1962) and a very slow decomposition rate (Melillo, Aber & Muratore, 1982). Bouliane (1962) experimentally showed that development and survival of maple seedlings is highly compromised when grown on beech humus. Shade is also a stress in the understory of mature forests. Even shade-tolerant species like maple require an opening in the early stages in order for the stem to be released. Difference in light level may contribute to the weakness of maple regeneration in the study site. We are currently conducting a field experiment to investigate the relative effect of light levels and qualities, as well as that of soil properties on maple regeneration in the Muirs' forest.

Beech is known to produce stems both from seedlings and root sprouts under closed canopy (Burns & Honkala, 1990). However, there is a great variation in their relative abundance according to geographical distribution (Ward, 1961; Held, 1983). Of the 112 beech stems we excavated in the Muirs' forest, 59 were from sprout origin, 20 from seed, and 33 of unknown origin. The uncertainty comes from the fact that the connection of the sprout with the parent tree may gradually be broken. The estimated 54% for root sprouts is thus conservative. There is some debate regarding the ability of root sprouts to develop into canopy trees (Burns & Honkala, 1990). Woods (1979, 1984) described a canopy-understory interaction where maple was thriving better under beech and, conversely, called this process "reciprocal replacement". However, in both of his studies, Woods (1979, 1984) excluded beech sprouts from the analysis, assuming that their survival was compromised and that they were unlikely to enter the canopy. They could be killed or damaged by the parent fall, or infected by disease through the connection. Cypher & Boucher (1982), who studied sapling growth of beech and maple under their canopy, rejected sprouts on the basis of the same arguments. On the other hand, evidence of beech sprouts reaching the canopy is, in some cases, indisputable. In some areas, the maintenance of beech could only be the consequence of the growth of root sprouts into canopy trees, since stems from seed origin are rarely or never found: this situation prevails in the Great Smokie Mountains (Ward, 1961) and in the Catskills (McIntosh, 1972). Moreover, Jones & Raynal (1987) found that low vigor trees produced healthier sprouts than did robust ones. In the Muirs' forest, many examples of canopy trees from sprout origin were found, some of which were still attached to the old decaying stump of the

parent tree. We believe that root sprouts contribute highly to the persistence and success of beech in the study site, as suggested by the strong self-replacement tendency (Table I). A similar view has been adopted elsewhere (Lauferweiler, 1955; Held, 1983).

The increase in hemlock, as suggested by the models, is also in accordance with its recognized successional status. Hemlock is shade tolerant (Burns & Honkala, 1990), and stems of small size have a low mortality rate (Rogers, 1978). Like beech, hemlock would affect its competitors by producing an unfavorable litter, and also by its deep shading (Daubenmire, 1931; Rogers, 1978). However, seed germination and seedling establishment are rare, possibly explaining its minor increase.

Basswood seedlings and saplings are rare in the mesic part of the forest, and the maintenance of this species appears to depend largely on vegetative resprouting. Because an extensive root structure already exists, a basswood sprout has a higher probability of replacing its parent tree than more abundant and shade-tolerant beech and maple saplings. This ability of basswood to maintain itself on mesic sites by vegetative resprouting rather than by seedlings has been widely recognized (Burns & Honkala, 1990; Woods & Whittaker, 1981).

At the northern limit of the Deciduous Forest in Québec, sugar maple has been considered the dominant species of the steady-state forest (Dansereau 1946, 1959; Grandtner, 1966). However, at the time of the first ecological surveys, the St. Lawrence Valley had already been extensively exploited for timber or deforested for agricultural use. The oldest forest stands were probably less than 100 to 150 years of age, which is young considering the life-span of the tree species involved. Because these immature forests seemed stable when their understory and overstory compositions were compared, a climax type was described accordingly. A comparison with the presettlement forest was not possible, for the land survey approach used under the French regime precludes its reconstruction, as in the United States (Bouchard *et al.*, 1989). In this context, an increase in beech over maple has been interpreted as an artificial degradation of the maple climax forest caused by human disturbance (Bouliane, 1962; Grandtner, 1966). The observed natural shift in the old-growth presettlement Muirs' forest nevertheless suggests that this view needs to be re-examined.

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